

Wearables and Habits: Synthetic Control Estimates

Issue #9

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1 Summary

Exploratory unit-level synthetic control of first wearable purchase on running-related purchases, as an alternative to the staggered DiD of #7 (which fails on pre-trends). Each adopter gets its own synthetic twin from never-treated users, matched on the 8 pre-adoption quarters; gaps are averaged over adopters in event time.

- **Pre-fit is easy, and removes the pre-trend by construction.** Pre-period gaps are about zero for every outcome (Figures 1, 2), so the ramp-up into adoption that broke DiD is absorbed by the synthetic twin.
- **Post gaps are positive, not specific to running.** Matching on the outcome’s own history, the post ATT (quarters 1–8) is +0.017 (0.006) for running shoes, +0.070 (0.012) running-related, +0.033 (0.009) fitness gear, +0.100 (0.016) supplements. But the placebo outcomes are positive too (pet +0.036, books +0.057, grocery +0.039) and so is total purchasing, $\ln(1 + n)$: +0.176 (0.038). Adopters simply buy more of everything afterwards than their synthetic twins.
- **Adding $\ln(1 + n)$ to the match shrinks the effects** (Table 2): running shoe +0.011 (0.006), running-related +0.043 (0.012), supplements +0.068 (0.017), and the placebos fall to +0.018, +0.030, -0.001 . Running-related and supplements remain above the placebos, but this is suggestive at best.
- **Not window-end attrition.** Requiring donors to be observed through the entire post window barely changes anything (e.g. running-related +0.066 own, +0.039 with $\ln n$ match; `synth_summary_fulldonor.csv`).
- **Reading.** The synthetic estimates and the #7 DiD disagree in sign, so neither is credible alone. A plausible driver of the common positive gap is selection of adopters on a purchase in $e = 0$ (and on ≥ 12 months of activity after), which donors are not selected on; this predicts positive gaps for all outcomes, as observed. The placebo test with pseudo-adopters drawn from donors (red) is ≈ 0 , but those units lack that selection, so it does not rule this out.

2 Design

- **Data:** the #7 user-quarter panel (`design_panel_q.csv`), “clean” sample. Treated: clean adopters with 8 observed pre-quarters, i.e. adoption in 2020Q1 or later (288 of 447). Donors: never-treated users (about 4,100) observed in all 8 pre-quarters.
- **Weights:** per adopter, non-negative weights summing to one on the 150 nearest donors (pre-period distance), least squares on the 8 pre-quarters (SLSQP). Spec “own” matches the

outcome only; “own+ln n ” also matches $\ln(1 + n)$. Outcomes: any purchase in the quarter (as in #7).

- **Effects:** $\text{gap}_{i,e}$ = adopter minus synthetic, $e = -8, \dots, 8$, $e = 0$ the adoption quarter; ATT_e = mean over adopters, SE = cross-adopter $\text{SD}/\sqrt{n}$ (ignores weight estimation error). Post ATT = mean of $e = 1, \dots, 8$ (excludes the device quarter).
- **Placebo:** 265 never-treated users get randomly drawn real adoption quarters and are run through the same procedure against the remaining donors (Placebo column, red points).
- Code: `code/scripts/synth_control.py`, `synth_control_figs.py`.

3 Results

Table 1: Synthetic control, matching on own pre-history. Pre RMSE = adopter-vs-synthetic RMS gap in pre-period; second figure is the RMS outcome level.

Outcome	Post ATT (SE)	Placebo (SE)	N	Pre RMSE / level
Running shoe	0.017 (0.006)	0.001 (0.005)	288	0.009 / 0.086
Running-related (keyword)	0.070 (0.012)	0.004 (0.009)	288	0.011 / 0.277
Fitness gear	0.033 (0.009)	-0.017 (0.006)	288	0.005 / 0.206
Supplements	0.100 (0.016)	-0.006 (0.015)	288	0.004 / 0.375
Medication	0.014 (0.013)	0.002 (0.012)	288	0.006 / 0.218
PLACEBO: pet	0.036 (0.016)	-0.013 (0.015)	288	0.006 / 0.396
PLACEBO: books	0.057 (0.015)	0.017 (0.016)	288	0.001 / 0.586
PLACEBO: grocery/food	0.039 (0.015)	-0.016 (0.015)	288	0.006 / 0.294
General: $\ln(1 + \# \text{purchases})$	0.176 (0.038)	-0.032 (0.044)	288	0.065 / 2.992

Table 2: Synthetic control, matching on own pre-history and $\ln(1 + n)$.

Outcome	Post ATT (SE)	Placebo (SE)	N	Pre RMSE / level
Running shoe	0.011 (0.006)	0.007 (0.005)	288	0.042 / 0.086
Running-related (keyword)	0.043 (0.012)	0.014 (0.009)	288	0.100 / 0.277
Fitness gear	0.021 (0.009)	-0.009 (0.006)	288	0.083 / 0.206
Supplements	0.068 (0.017)	-0.010 (0.014)	288	0.111 / 0.375
Medication	-0.009 (0.012)	0.015 (0.012)	288	0.076 / 0.218
PLACEBO: pet	0.018 (0.016)	-0.010 (0.015)	288	0.100 / 0.396
PLACEBO: books	0.030 (0.015)	0.010 (0.015)	288	0.138 / 0.586
PLACEBO: grocery/food	-0.001 (0.015)	-0.021 (0.014)	288	0.095 / 0.294

With the own-only match, pre-fit RMSE is 0.001–0.015, an order of magnitude below the outcome level, which suggests interpolation (8 targets, 150 donors); the $\ln n$ match fits less tightly (RMSE 0.04–0.15) and is the more conservative specification (`synth_es_main_own.png`, `synth_es_placebo_own.png`).

4 Caveats and follow-ups

- Treated sample is the later adopters only (2020+), so not comparable to #7’s 447.
- Follow-ups (not done): hold-out pre-period fit to test overfitting; synthetic DiD / augmented SC with a unit fixed effect; pseudo-adopters selected on an $e = 0$ purchase to test the selection

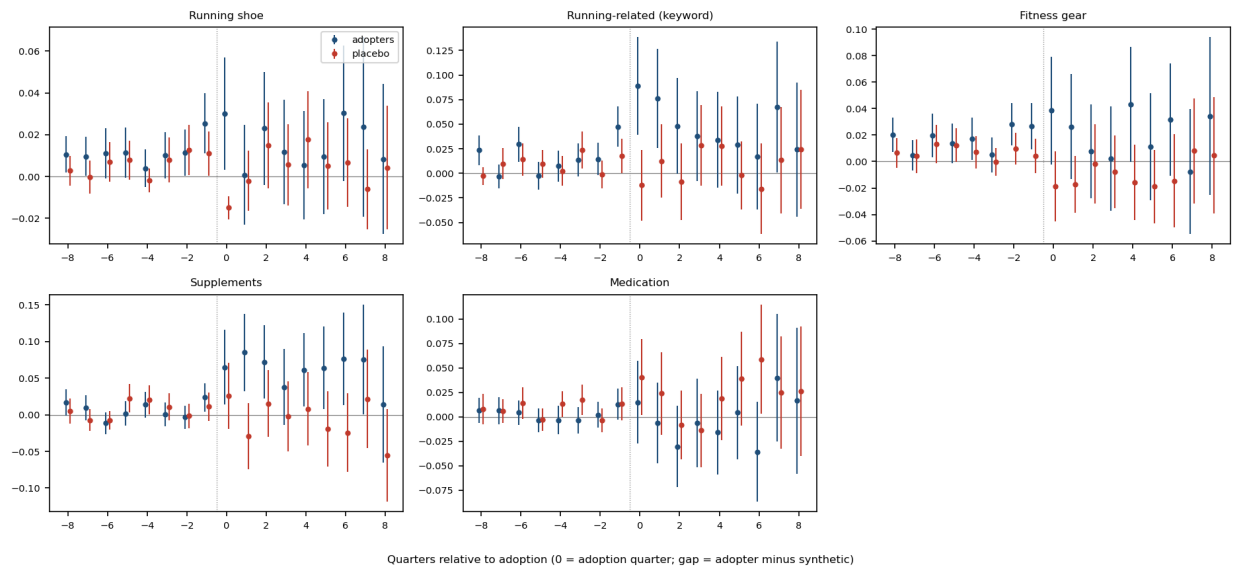


Figure 1: Event-study gaps, main outcomes, matching on own history and $\ln n$ (blue adopters, red placebo; 95% CIs).

explanation; not-yet-treated donors; HonestDiD on #7.

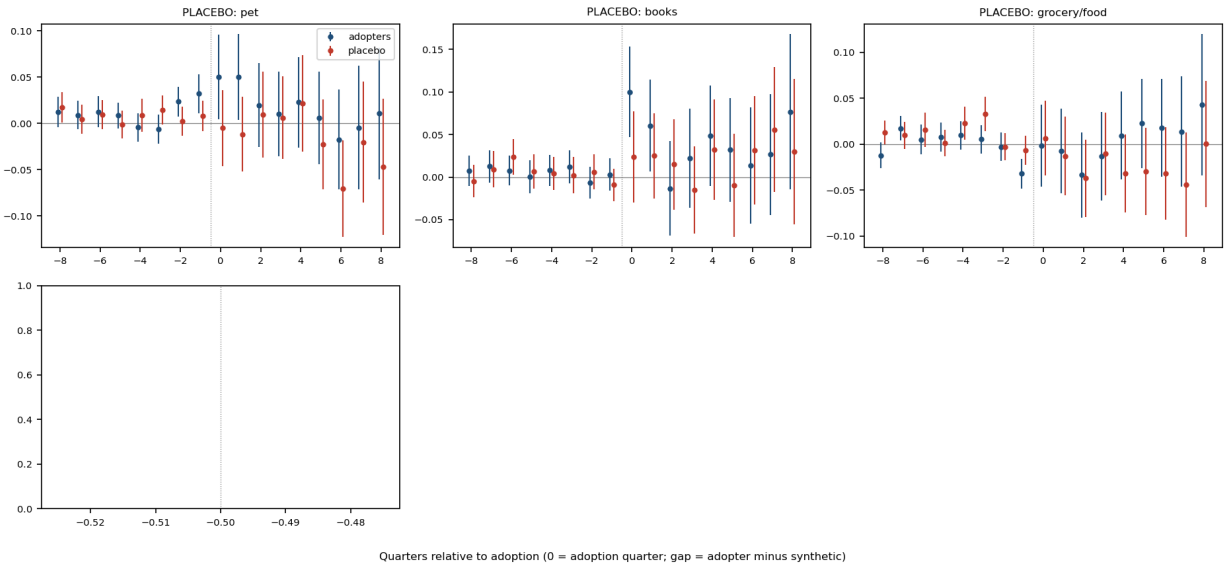


Figure 2: Same, placebo outcomes.